

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) steers generative training by distilling representational structure from a powerful pretrained vision encoder into intermediate diffusion features. We examine a fundamental question: which component of the target representation is consequential for generation—its *global* semantic information (e.g., as reflected by ImageNet-1K accuracy) or its spatial organization (i.e., the pairwise cosine-similarity geometry among patch tokens)? Although the prevailing interpretation suggests that stronger global semantic performance should yield a superior target representation for generation, our large-scale empirical analysis, spanning 27 vision encoders and multiple model scales, reveals a markedly different pattern: spatial structure, rather than global performance, is the dominant predictor of generative efficacy. To interrogate this observation further, we introduce two deliberately simple modifications that selectively amplify the transfer of *spatial* information. Specifically, we replace the conventional MLP projection layer in REPA with a lightweight convolutional layer and apply a spatial normalization layer to the external representation. Despite requiring fewer than four lines of code, our method, termed iREPA, consistently accelerates REPA convergence across diverse vision encoders, model sizes, and training variants, including REPA, REPA-E, Meanflow, and JiT. Collectively, these results motivate a reconsideration of the operative mechanism underlying representational alignment and of the ways in which it can be exploited for more efficient generative-model training.

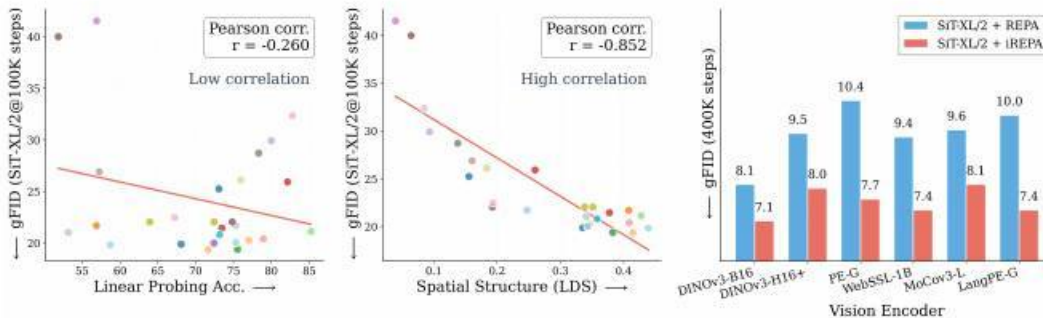


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Contrary to prevailing assumptions, spatial structure, rather than global performance measured by linear-probing accuracy, is the representational property most strongly associated with target-representation generation performance. *Right:* We further investigate this mechanism by introducing two simple modifications that accentuate the transfer of spatial features from the target representation to the diffusion model. This simple approach consistently improves REPA convergence speed across diverse settings.

1 INTRODUCTION

Representation alignment has recently become a particularly effective mechanism for accelerating the optimization of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By constraining



Figure 2: **Motivating examples: spatial structure matters.** Metric comparisons reveal an inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite substantially higher validation accuracy (82.8% vs. 53.1%), exhibits worse generation quality than SpatialPE-B (Bolya et al., 2025). *Right:* Similarly, WebSSL-1B (Fan et al., 2025) has much stronger global performance (76.0% vs. 53.1%) but worse generation. **Spatial Self-Similarity:** Spatial structure provides a better predictor of generation quality than global performance. See §3 for the spatial-structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

internal diffusion features to align with representations produced by pretrained self-supervised visual encoders, recent methods have achieved substantial improvements in both convergence rate and eventual sample quality (Yu et al., 2024; Leng et al., 2025a). Yet, despite these empirical gains, the precise mechanism through which self-supervised features improve diffusion training remains insufficiently characterized. A central question therefore remains unresolved: are the benefits primarily attributable to the injection of stronger global semantic information, as is commonly estimated through linear-probing performance, or do they instead arise from the preservation and transfer of spatial structure, namely the relational geometry among patch-token representations?

Clarifying this mechanism is consequential for generative-model training because it determines how one should select a target representation and how one should maximize the benefit obtained from alignment. The dominant assumption is that the usefulness of an encoder for representation alignment should correlate with ImageNet-1K validation accuracy, which is often treated as a proxy for global semantic competence (Oquab et al., 2024; Chen et al., 2021). Equivalently, target representations that perform better on ImageNet are hypothesized to produce better generative models when used within REPA (Yu et al., 2024). Consequently, improvements in the linear-probing accuracy of diffusion features are frequently invoked as evidence for the efficacy of representation alignment, thereby reinforcing the view that global semantic information is the principal source of improvement. The following statement from REPA (Yu et al., 2024) succinctly captures this interpretation:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we revisit this conventional account by systematically asking what actually governs the effectiveness of representation alignment: global semantic information or spatial structure? Through a large-scale empirical study over heterogeneous vision encoders—including recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)—we identify three findings that materially revise the standard interpretation.

Higher validation accuracy does not imply better representation for generation. Contrary to the prevailing assumption, vision encoders with stronger global semantic performance, as measured by ImageNet-1K linear-probing accuracy, can perform worse when used as target representations for generation. For example, PE-Spatial-B, a spatially tuned model derived from PE-Core-G (Bolya et al., 2025), has substantially lower patch-token validation accuracy¹ than PE-Core-G (53.1% vs. 82.8%), yet yields better generation under REPA (Figure 2). Analogously, WebSSL-1B (Fan et al., 2025) exhibits much stronger global performance (76.0% vs. 53.1%) while producing inferior generative results. In Section §2, we show that this pattern persists across target representations, indicating that the mismatch between semantic validation accuracy and generative utility reflects a broader principle rather than an isolated anomaly.

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

Spatial structure is a better predictor of generation performance. To formalize this alternative explanation, we quantify the spatial organization of target representations through spatial self-similarity metrics (Shechtman & Irani, 2007), which measure how patch-token similarity varies as a function of spatial distance (§3). Across 27 diverse vision encoders, these spatial metrics correlate much more strongly with generation performance than ImageNet-1K accuracy does (Pearson $|r| > 0.85$ vs. $|r| = 0.26$). Moreover, spatial structure explains several otherwise puzzling observations (Figure 3): SAM2-S, despite its comparatively weak ImageNet-1K accuracy, works well for REPA because its representations retain strong spatial organization; larger encoders within the same family can underperform when their spatial structure deteriorates; and injecting global information from CLS tokens into patch tokens improves linear probing while degrading generation, because it suppresses spatial contrast. This interpretation is further supported by experiments with classical spatial descriptors, including SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), which still provide nontrivial gains when used for representation alignment (§3).

Accentuating spatial transfer improves REPA. Motivated by these observations, we introduce iREPA, a minimal modification of REPA that strengthens the transmission of spatial structure from the target representation to the diffusion transformer. iREPA replaces the MLP projection layer used in REPA with a lightweight convolutional projection and applies spatial normalization to the external representation, as visualized in Figure 6. These changes are intentionally simple, yet they consistently improve convergence across vision encoders, diffusion-model scales, and training variants, including REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025). Our results therefore suggest that representation alignment should be understood less as the transfer of global semantic category information and more as a mechanism for imposing spatially structured representational geometry during generative training.

In summary, our contributions are as follows:

- We provide a systematic empirical analysis showing that global semantic performance, as measured by ImageNet-1K linear probing, is not a reliable predictor of a target representation’s utility for generation.
- We introduce spatial self-similarity metrics and show that they correlate substantially more strongly with REPA generation performance across encoders and model scales.
- We propose iREPA, a simple spatially oriented modification of REPA that improves convergence across encoders, model sizes, and training recipes, including REPA, REPA-E, and MeanFlow with REPA.

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with several motivating examples demonstrating that higher global performance in a target representation, operationalized through ImageNet-1K accuracy, does not necessarily translate into stronger generation performance under REPA. We subsequently show in §3 that these otherwise anomalous observations are more coherently explained by measuring the spatial structure of the target representation.

Recent vision encoders. We examine a diverse set of recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As shown in Figure 2, PE-Core-G attains substantially higher ImageNet-1K accuracy (82.8% vs 53.1%), yet it performs worse when used as the target representation for REPA (FID 32.3 vs 21.0). A comparable discrepancy appears for WebSSL-1B (1.2B), which achieves much higher ImageNet-1K accuracy than PE-Spatial-B (76.0% vs 53.1%) but nevertheless yields inferior REPA generation performance (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To further assess the limited explanatory power of global information for generation, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024), a small model with minimal global information and a validation accuracy of only 24.1% (Fig. 3a). Despite this weak global score, SAM2-S achieves better FID under REPA than several encoders whose ImageNet-1K accuracies are dramatically higher, including PE-Core-G (82.8%).

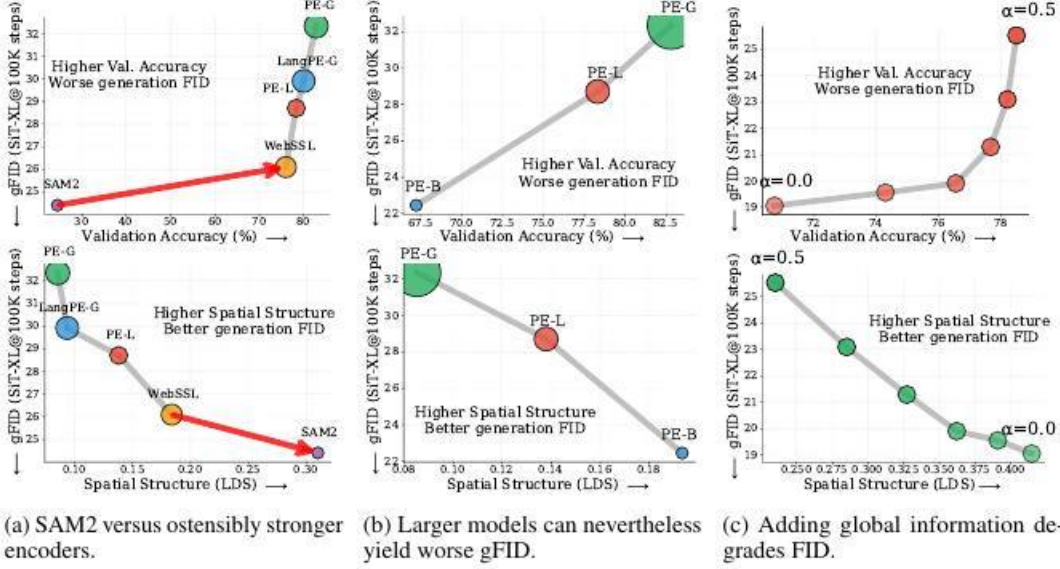


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Several trends indicate that global performance is poorly aligned with generation FID under REPA. (a) SAM2-S, despite a validation accuracy of only 24.7%, achieves better REPA generation performance than models with approximately $\sim 60\%$ higher validation accuracy. (b) Larger encoders within the same family can improve validation accuracy while degrading generation performance. (c) Adding global information to patch tokens through the CLS token improves global performance but harms generation. **Bottom row:** Spatial structure, rather than global performance, provides a more informative indicator of generation quality. See §3 for a large-scale analysis across spatial-structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

Larger models within same encoder family can have similar or worse generation performance. A natural expectation is that larger variants within a given encoder family should provide superior target representations, especially when this superiority is measured by ImageNet-1K accuracy. For representation alignment, however, larger variants frequently produce similar (DINOv2) or even *worse* (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a related observation for DINOv2 and attribute it to the possibility that all DINOv2 models, having been distilled from DINOv2-g, share similar representations. We later show that the same trend admits a more direct explanation in terms of spatial structure (§3).

Adding more global information can hurt generation. To test whether additional global information benefits representation alignment, we conduct controlled experiments in which global semantics from the CLS token are injected into local patch tokens, using DINOv2 as the encoder. The CLS-token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ denotes the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. As Figure 3c shows, increasing α from 0 to 0.5 monotonically improves linear-probing accuracy from 70.7% to 78.5%. Nevertheless, generation quality deteriorates substantially, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

Taken together, these observations indicate that the global performance of a representation is an unreliable proxy for its usefulness within REPA. We next demonstrate (§3) that spatial structure provides a more predictive and mechanistically informative signal.

- **Finding 1.** Better global semantic information (e.g., ImageNet-1K accuracy) does not imply a better representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We hypothesize that the generative utility of a target representation is governed more by its spatial structure than by its global information. To quantify this property, we first define several simple metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then show that these spatial-structure metrics not only correlate far more strongly

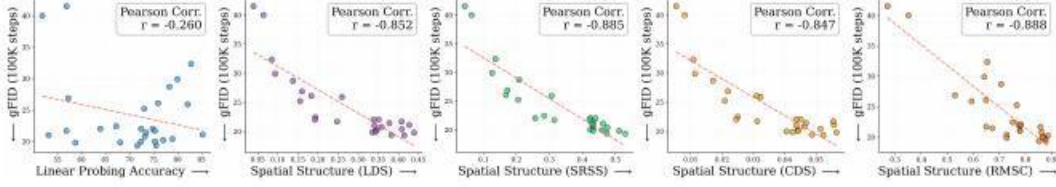


Figure 4: Spatial structure correlates more strongly with generation quality than linear probing. Correlation analysis across 27 diverse vision encoders using SiT-XL/2 and REPA. Linear probing exhibits weak correlation with FID (Pearson $|r| = 0.260$), whereas all spatial-structure metrics—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—show substantially stronger correlation with generation performance. See Fig. 10 for detailed plots with encoder labels.

with gFID than ImageNet-1K accuracy does, but also explain the previously puzzling phenomena described in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ denote the extracted patch representations, with $T = H \times W$ patches. Let the kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. In our experiments, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ specify the coordinate-space location of each of the T tokens, and let $d(\cdot, \cdot) \in \mathbb{N}$ denote the Manhattan distance between token pairs.

We then define a *spatial* self-similarity metric by measuring how the self-similarity kernel $K_{\mathcal{X}}$ changes as a function of the lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization, because nearby patches are more similar to one another than to distant patches. Unless otherwise stated, we use a simple correlogram-contrast metric contrasting local and distant similarities (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed by taking the expectation over the patch representation $\mathcal{X}$. We set $r_{\text{near}} = r_{\text{far}} = H/2$ in the main experiments, although we find that the resulting correlations are robust to these precise choices. Appendix B provides exact implementation details and alternative spatial metrics (CDS, SRSS, RMSC), which also perform effectively (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing.

We next conduct a large-scale correlation analysis across 27 heterogeneous vision encoders. As shown in Figure 4, the commonly used linear-probing measure exhibits only weak correlation with generation performance (Pearson $|r| = 0.26$), whereas all SSM metrics display substantially stronger associations (Pearson $|r| > 0.85$).

Generalization across model scales. We further evaluate whether these correlations persist across model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing remains weakly correlated at every scale ($|r| < 0.306$), while spatial structure consistently maintains a much stronger relationship with generation performance ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates much higher with generation performance than linear probing. We next use spatial metrics to explain previously unexplained trends.*

Spatial structure can explain previously unexplained trends. As established in §2, global performance measured by ImageNet-1K accuracy is not a reliable predictor of a representation’s effectiveness for alignment-based generation. The spatial metrics introduced above provide a more coherent explanatory account.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows that PE-Core-G achieves much higher ImageNet-1K accuracy than PE-Spatial-B (82.8% vs 53.1%) but performs worse as the REPA target representation (FID 32.3 vs 22.0). The spatial-structure metric clarifies this discrepancy: despite its weaker global performance, PE-Spatial-B exhibits substantially stronger spatial structure than PE-Core-G, thereby yielding the observed improvement in generation performance (see also Figure 2).

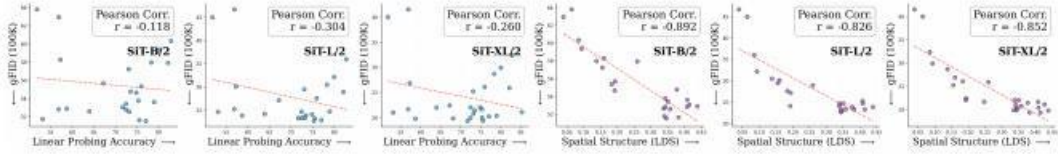


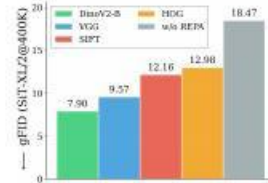
Figure 5: **Correlation analysis across model scales.** Across model scales, spatial structure (right) consistently exhibits stronger correlation with gFID than linear probing (left).

(2) *SAM2 outperforms “better” vision encoders:* §2 shows that SAM2, despite its much lower ImageNet-1K accuracy (24.1%), produces better generation than encoders with approximately $\sim 60\%$ higher accuracy. As illustrated in Fig. 3a, these gains are directly explained by SAM2’s stronger spatial structure.

(3) *Larger models in same encoder family underperform:* As shown in Fig. 3b, larger models within the same encoder family often improve ImageNet-1K accuracy while exhibiting weaker spatial structure, which in turn leads to poorer generation under REPA. This interpretation is consistent with observations in Yu et al. (2024), where larger models are reported to achieve similar or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation:* In Figure 3c, increasing global information by mixing the CLS token into patch tokens improves global performance. However, CLS-token mixing reduces spatial contrast among patch tokens, causing otherwise unrelated regions (e.g., foreground objects and background areas) to become excessively similar. The resulting attenuation of spatial structure consequently degrades generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Surprisingly, yes. Although they are not the strongest alternatives, classical spatial descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) nevertheless produce performance gains under REPA. This observation provides additional evidence that representation alignment can benefit from spatially structured features even in the absence of rich global semantic information.



Can we use spatial metrics to explain gains with REPA? Yes. As shown, the proposed spatial metrics explain both the improvements obtained by REPA and the gains produced by our improved training recipe, iREPA, which we introduce next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

To further interrogate the role of spatial structure in representation alignment, we introduce two deliberately minimal modifications to the original REPA training recipe, each intended to intensify the transmission of spatially organized information from the teacher representation, supplied by the vision encoder, to the student representation learned by the diffusion transformer.

Convolutional projection layer instead of MLP. The standard REPA formulation employs a three-layer MLP projection to map diffusion-feature dimensions into the dimensionality of the external representation; however, as shown in Figure 6a, this projection is lossy and attenuates the spatial contrast between patch tokens. We therefore replace the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly on the spatial grid and, by retaining a spatially local inductive bias, more naturally preserves neighborhood-level relationships among tokens.

Spatial normalization layer. Consistent with Siméoni et al. (2025), we find that patch tokens produced by pretrained vision encoders contain a substantial global component, as indicated by the high linear-probing scores obtained from the mean of patch tokens (Figure 14). Figure 6b further shows that, although mixing this global information (the mean of patch tokens) into the individual patch tokens improves global performance, it simultaneously diminishes the spatial contrast among those tokens. Consequently, tokens associated with semantically or spatially distinct regions (e.g., foreground objects and background regions) can acquire inappropriately high similarity.

Given the results in §3, we hypothesize that sacrificing this global component (the mean of patch tokens) can increase the spatial contrast among patch tokens. This enhanced contrast should provide a stronger spatial signal, expressed through pairwise patch-token similarity, and thereby improve REPA performance. To this end, we apply a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ ensures numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

We validate the resulting training recipe through extensive ImageNet 256×256 experiments, structuring the evaluation around three questions that probe robustness across target representations, model scales, implementation choices, and training recipes:

- Does iREPA consistently accelerate the convergence of diffusion transformers relative to REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Does iREPA scale with model size and remain effective under variations in the training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA transfer to more recent representation-alignment methods, such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025), as well as to pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across multiple vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). As reported in Fig. 7, iREPA consistently attains stronger performance at earlier training stages than baseline REPA, indicating that the proposed spatial modifications remain beneficial across variation in both the target representation and the diffusion-model scale.

Target representation. We analyze the generalization of iREPA across different vision encoders in Fig. 8 and Table 4. Across the evaluated encoders, iREPA consistently improves generation quality, and additional comparisons over all 27 encoders are provided in Appendix C, E.

Enc. Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑	Model Size	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑	Aln. Depth	IS↑	FID↓	sFID↓	Pr.↑	Rc.↑
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

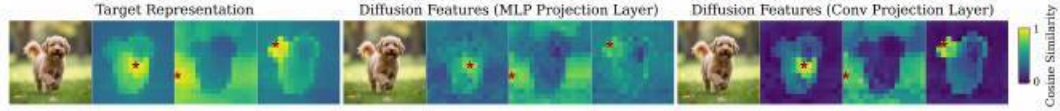
(a) Vision-encoder size

(b) Model size

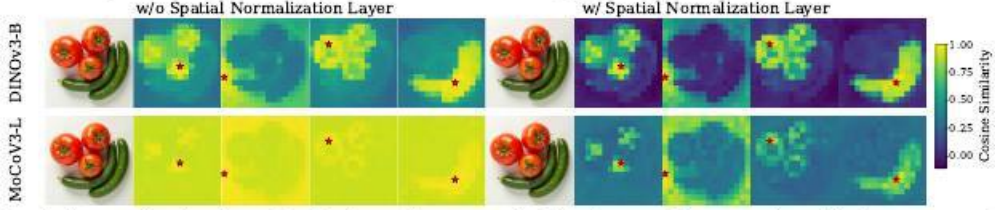
(c) Encoder depth (SiT-B/2)

Table 1: Variation in training settings. iREPA yields consistent gains over baseline REPA across diverse training settings. Unless otherwise specified, all results use Dinov2 as the encoder, SiT-XL/2, 100K steps, and vanilla REPA as the baseline.

Encoder size. We examine the generalization of iREPA across encoder sizes in Table 1a, using PE-B (90M), PE-L (320M), and PE-G (1.88B). iREPA improves performance for every encoder size;



(a) *Simpler projection layer for REPA.* Standard MLP projection layer in REPA (middle) loses spatial information while transferring features from target representation (left) to diffusion features. Instead using a simpler convolution layer leads to better spatial information transfer (right).



(b) *Spatial normalization layer.* Patch tokens from pretrained vision encoders contain a global component that limits spatial contrast. Consequently, tokens in one semantic region (e.g., tomato) can exhibit substantial cosine similarity with unrelated tokens (e.g., background or cucumber). We hypothesize that sacrificing this global component can increase patch-token spatial contrast and thereby improve generation performance.



(c) *Overall impact* of the improved training recipe (iREPA) on the spatial structure of diffusion features.

Figure 6: Two straightforward iREPA modifications designed to enhance spatial-feature transfer from the target representation to diffusion features. All results are reported with SiTL/2 using REPA at 100K.

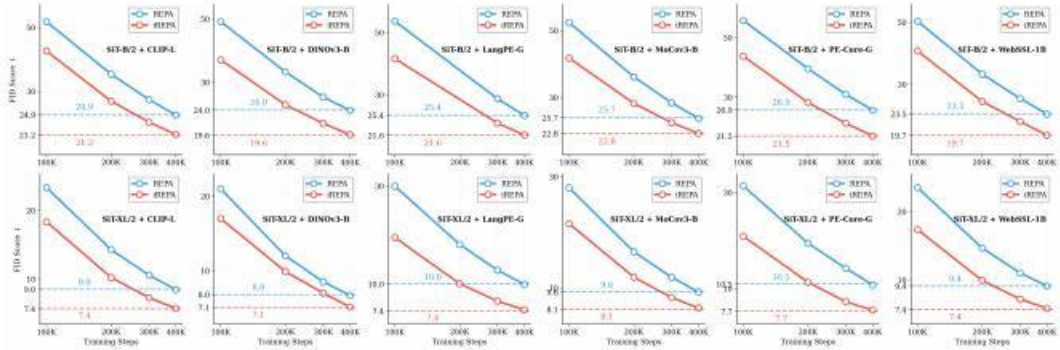


Figure 7: **Accentuating spatial features consistently improves convergence speed.**

notably, the relative gain increases with encoder scale, rising from 22.2% for PE-B to 38.8% for PE-L and 39.6% for PE-G.

Scalability. We evaluate the scalability of iREPA across diffusion-model scales in Table 1b. The spatial modifications consistently improve performance and yield larger percentage gains for larger models, suggesting that the benefits of spatially oriented alignment scale favorably with model capacity.

Encoder depth. Table 1c examines whether iREPA generalizes across alignment depths, using SiT-B/2 at 100K iterations with DINOv3-B as the target representation. Across these depths, iREPA delivers consistent improvements over baseline REPA.

Ablation on different components. We additionally isolate the contribution of each iREPA component in Table 2. Both spatial normalization and the convolutional projection layer substantially improve generation quality relative to baseline REPA, with the strongest results obtained when the two components are used together.

Training recipe. Finally, we test whether iREPA generalizes across training recipes, including REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). Table 3 shows that the spatial

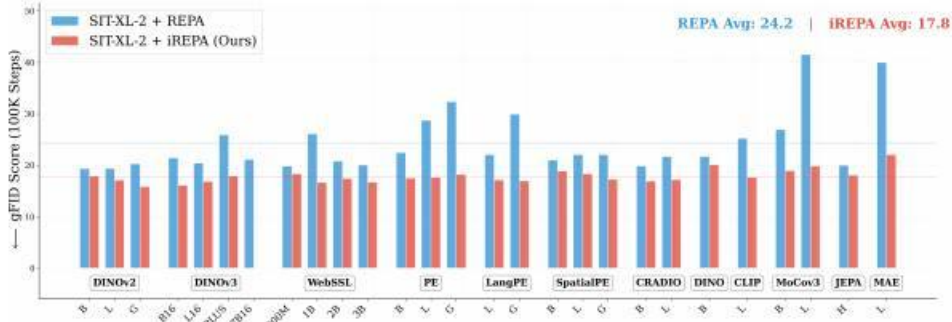


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from the target representation to diffusion features with iREPA yields consistent convergence-speed improvements. See Appendix C for additional results across diverse settings.

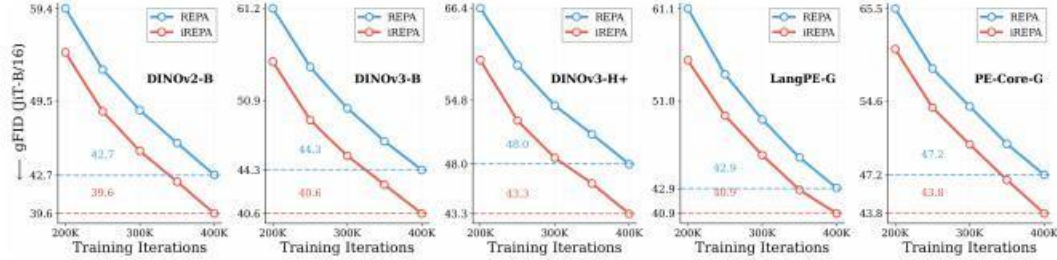


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves compare REPA and iREPA with pixel-space JiT (Li & He, 2025b) across different vision encoders. Accentuating the transfer of spatial structure consistently accelerates convergence across vision encoders for pixel-space diffusion models. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

modifications introduced by iREPA produce convergence gains across these distinct optimization recipes.

Classifier-free guidance. We evaluate iREPA under classifier-free guidance in Table 4, which reports results across several vision encoders at 400K training iterations. Across encoders, iREPA accelerates convergence irrespective of whether classifier-free guidance is used.

Pixel-space diffusion (JiT). We also evaluate iREPA on pixel-space diffusion models, specifically JiT-B (Li & He, 2025b). Figure 9 reports results for both REPA and iREPA within JiT (Li & He, 2025b), showing that an explicit emphasis on spatial-information transfer consistently accelerates REPA convergence across multiple vision encoders (e.g., DiNOv2, DiNOv3, PE *etc.*).

- **Finding 3.** Accentuating transfer of spatial structure from target representation to diffusion features helps boost convergence speed with representation alignment (REPA).

5 RELATED WORK

We summarize the most directly relevant bodies of work here and defer a more expansive discussion to Appendix H.

Representation alignment for generation. A growing body of recent work investigates the use of external representations to improve diffusion-model training (Pernias et al., 2023; Fuest et al., 2024). In particular, recent approaches (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) demonstrate that substantial performance gains can be obtained by aligning internal diffusion features with clean-image features extracted from a pretrained vision encoder. This idea has

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across different encoders, both the spatial-normalization layer and the convolutional projection layer (§4) yield substantial convergence-speed gains, with the best results obtained when both are used together. All results are reported at 100K steps with SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** We apply the proposed spatial improvements on top of REPA-E (a) and Meanflow with REPA (b), obtaining consistent gains. For Meanflow, we report results at 1 and 4 NFEs. A CFG value of 2.0 is used for Meanflow. All results are reported at 100K training iterations.

also been extended to video generation and 3D generation by (Zhang et al., 2025; Wu et al., 2025), respectively, while (Ma et al., 2025a) shows that representation alignment can improve the training of unified models. Despite these empirical successes, the mechanisms by which self-supervised features enhance diffusion training remain only partially understood. Our work addresses this gap by identifying which aspect of the target representation matters for generation and by using that insight to formulate an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. Recent studies examine the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued training of self-supervised vision representations can increase similarity between the global CLS token and patch tokens, thereby degrading performance on dense spatial tasks. Complementarily, (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatially tuned models for dense spatial prediction. In this paper, we show that, for generation, the spatial structure of a vision encoder is more consequential than its global information, a finding that we hope will motivate more principled criteria for selecting and training external representations for generative alignment.

6 CONCLUSION

In this paper, we investigated the mechanism underlying the effectiveness of representation alignment, asking whether its benefits are principally mediated by global information or by the spatial structure of the target representation. Through large-scale empirical analysis, we identified a counterintuitive but consistent pattern: spatial structure, rather than global information, is the dominant factor associated with effective representation alignment. We further examined this finding by introducing two simple modifications that accentuate the transfer of spatial information from the target representation to diffusion features. The resulting method, iREPA, consistently improves REPA convergence across diverse vision encoders, model scales, and training recipes. Taken together, these results suggest that future work should revisit the operative mechanism of representational alignment and develop more systematic ways of exploiting spatially structured representations for generative-model training.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+IREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+IREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+IREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+IREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+IREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+IREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+IREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+IREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across different encoders with and without classifier-free guidance.** SiT-XL/2 is used as the base model. See Appendix E for detailed SiT-B/2 and SiT-L/2 results across all encoders.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

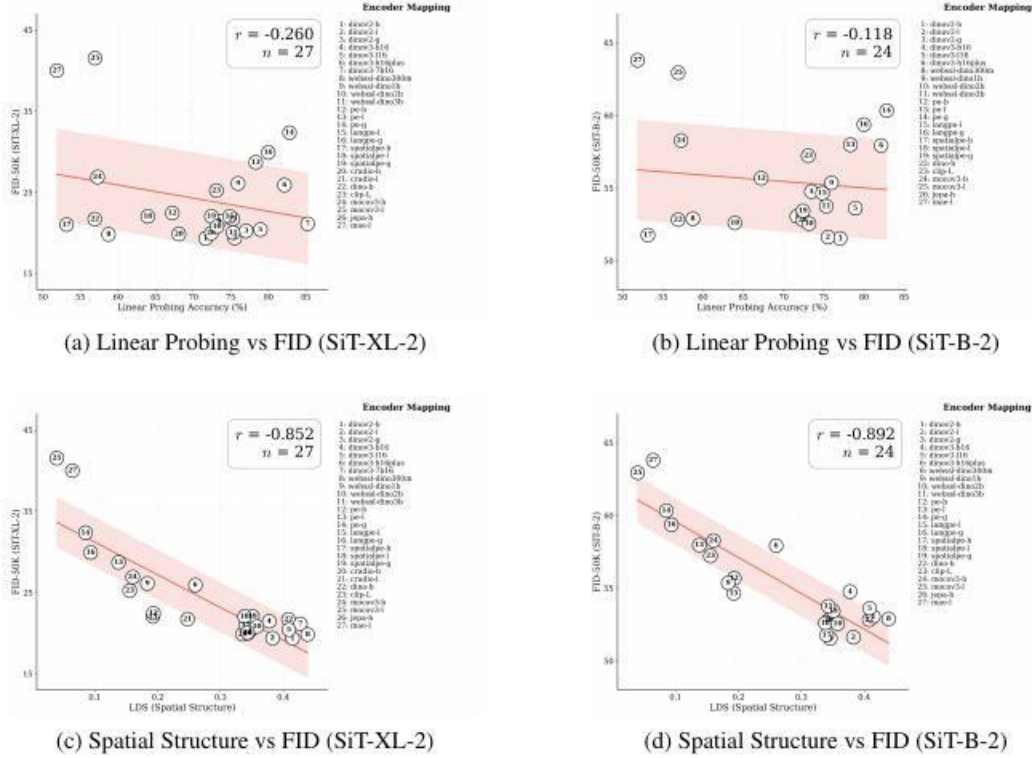


Figure 10: Detailed correlation analysis with encoder labels. The top row reports the relationship between linear-probing accuracy and FID for SiT-XL-2 and SiT-B-2 models, whereas the bottom row reports the corresponding relationship between spatial structure (LDS) and FID. Each point is annotated with its encoder identifier (see legend). Across both model scales, the spatial-structure metric exhibits a substantially stronger association with generation quality.

B SPATIAL SELF-SIMILARITY METRICS



Figure 11: **Explaining the semantic-region self-similarity (SRSS) metric.** Visual explanation of the SRSS metric: for each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. SSM measures the difference in cosine similarity between anchor-positive pairs within the mask and anchor-negative pairs outside the mask. Intuitively, larger values indicate that patches in the same semantic region are more similar than patches in unrelated regions, indicating better preservation of spatial structure.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, where $T = H \times W$ and token indices $t \in [T]$ are arranged on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ denote Manhattan distance, and let the cosine self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

which is a standard measure of local self-similarity (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, where $T = H \times W$ indexes the spatial grid of patches (patch index set $[T]$). We additionally let $d : [T] \times [T] \rightarrow \mathbb{N}$ denote Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measures self-similarity (Shechtman & Irani, 2007) between patch tokens. We instantiate this quantity using the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ quantifies how patch-token self-similarity K varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization, meaning that nearby patches are more similar to one another than to distant patches.

We next describe several direct metrics for quantifying the spatial self-similarity structure of target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}],$$

where r_{near} and r_{far} are hyperparameters. By default, we set $r_{\text{near}} = r_{\text{far}} = H/2$, and we find that the correlations are robust to the precise choices of these radii. Intuitively, larger values indicate that, on average, spatially proximate patches are more similar than patches separated by larger distances.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values correspond to faster decay of similarity with spatial distance, and therefore indicate stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first use SAM2 (Ravi et al., 2024) to sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object, and we downsample that mask to $H \times W$. We then select anchor, positive, and negative points: the anchor and positive points are sampled from within the mask, where they should share semantics, whereas the negative point is sampled outside the mask and is therefore less related to the anchor. If an encoder representation

preserves spatial structure, the anchor-positive cosine similarity should exceed the anchor-negative cosine similarity. Accordingly, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

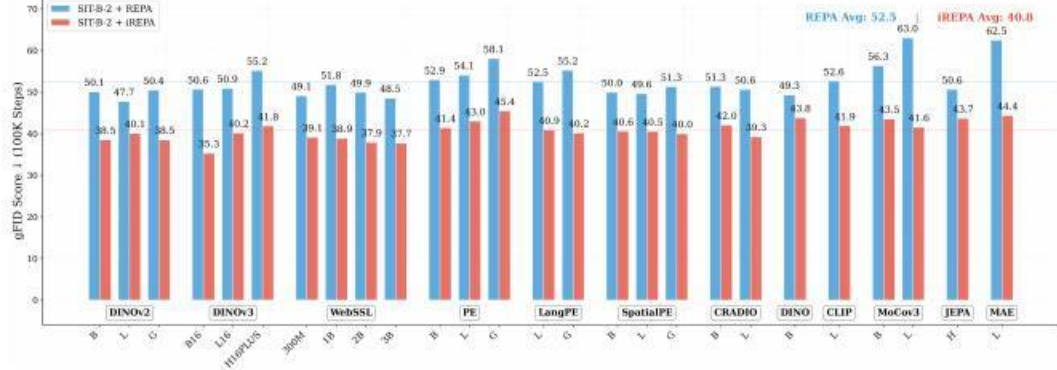
where positive points lie within Manhattan distance $d \leq r_{\text{near}}$ of the anchor inside the mask, while negative points lie at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric that captures the degree of spatial variation among patch-token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

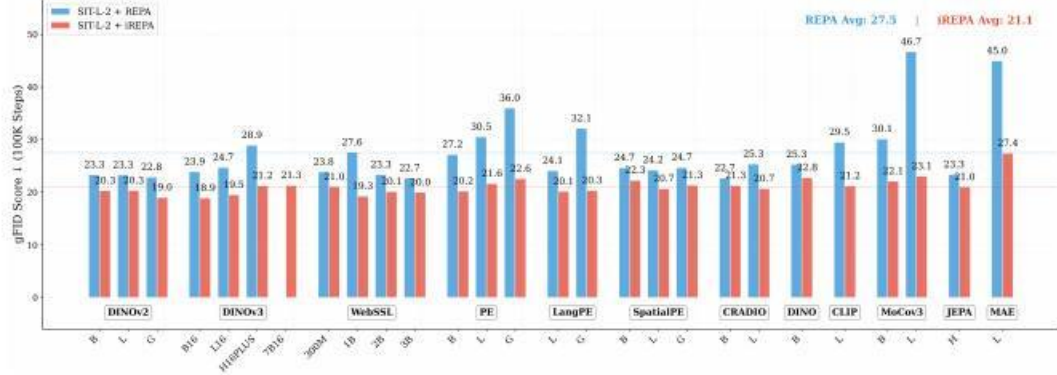
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean normalized feature across spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representation and hence better-preserved spatial structure, whereas lower values indicate more uniform feature distributions consistent with spatial-structure loss, which often occurs when global information diminishes spatial contrast (Siméoni et al., 2025)).

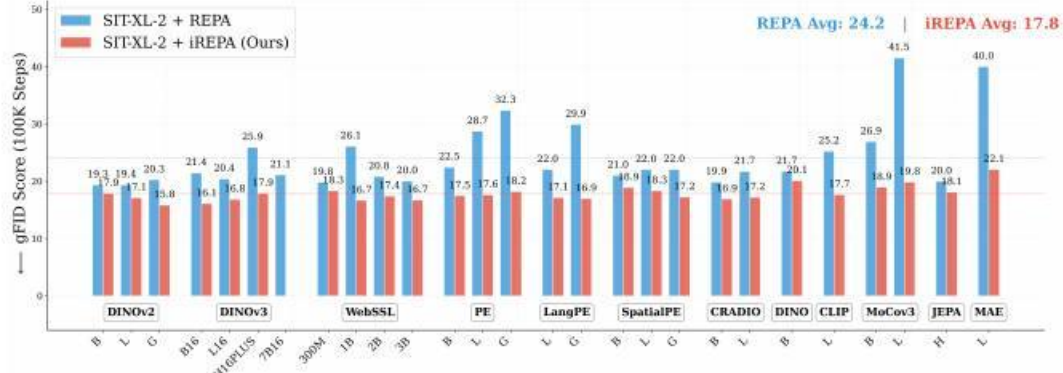
C VISION ENCODER VARIATION



(a) SiT-B-2 with 130M parameters



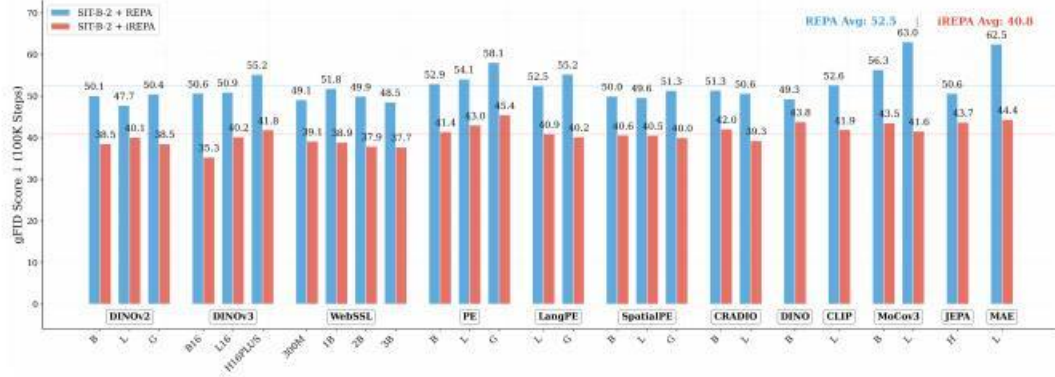
(b) SiT-L-2 with 458M parameters



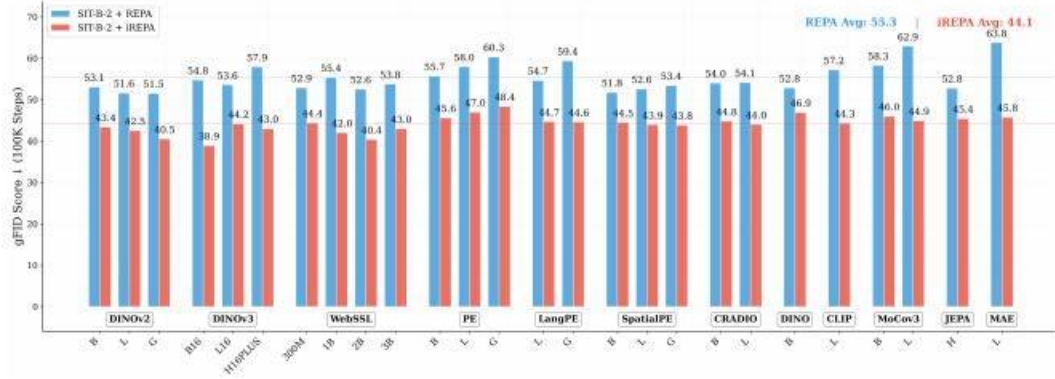
(c) SiT-XL-2 with 675M parameters

Figure 12: **Variation in vision encoders.** Ablation studies show iREPA improvements across different vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently improves generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across various vision encoders. The results show consistent improvements with iREPA regardless of encoder depth, with slightly better performance observed with 8 encoder layers. Both configurations demonstrate significant FID improvements when spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with particularly strong gains in FID and IS metrics. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

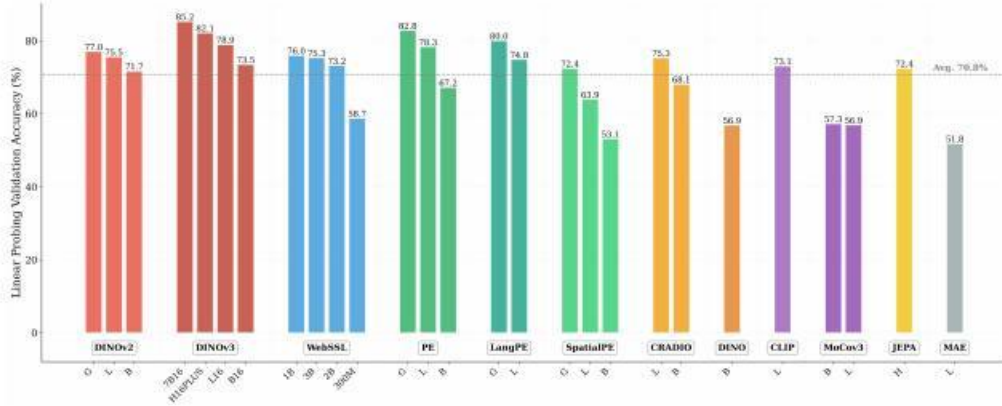


Figure 14: **Global information in mean patch tokens.** In addition to the [CLS] token, the mean of patch tokens also contains substantial global semantic information. Although this information improves global performance, it reduces contrast among individual patch tokens and can therefore hinder spatial-structure transfer. We find that removing part of this global information through the mean of patch tokens improves spatial-structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA and iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both IS and FID across all encoders.

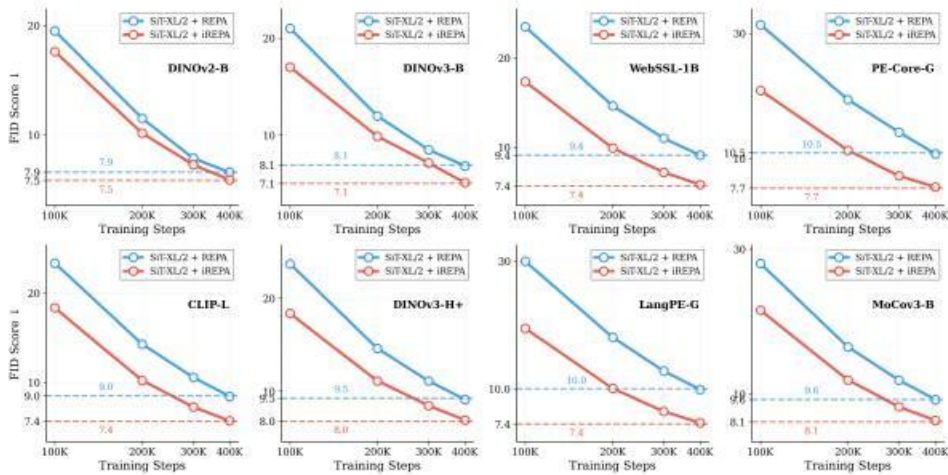


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

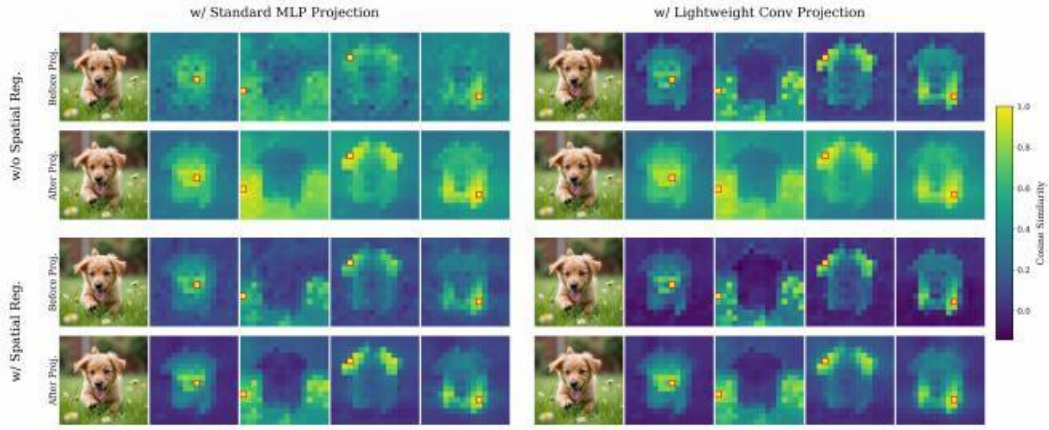


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, the standard MLP projection layer (top-left) loses spatial information while transferring features between the pretrained encoder (after projection) and the diffusion model (before projection). Using a simpler convolutional layer instead better preserves spatial-information transfer (top-right). **Second**, vision-encoder features often exhibit limited spatial contrast (§2). This causes tokens in one semantic region (*e.g.*, dog) to show substantial cosine similarity with unrelated tokens (*e.g.*, background). We address this issue with a simple spatial regularization layer that accentuates spatial contrast in the learned representation, thereby improving generation performance. The strongest results are obtained when both modifications are used together (see Table 2).

G IMPLEMENTATION DETAILS

Training setup. Unless otherwise specified, we follow the experimental setup of REPA (Yu et al., 2024). All training uses the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the ADM (Dhariwal & Nichol, 2021) protocol, under which each image is center-cropped and resized to 256×256 resolution. Throughout diffusion training and inference, we use the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.). For the spatial-normalization layer, we set $\gamma \in [0.6, 0.8]$; for the projection layer, we use a convolutional layer with kernel size 3 and padding 1. Optimization is performed with AdamW (Kingma & Ba, 2014; Loshchilov, 2017), a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` for acceleration, together with gradient clipping and an exponential moving average (EMA) of the generative model for stable optimization.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we use the implementation, including REPA support, provided by the original authors and adapt it by introducing the two straightforward iREPA modifications described in §4.

Evaluation. For image-generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.), and recall (Rec.) (Kynkäänniemi et al., 2019), all measured on 50K generated images. For sampling, we follow REPA (Yu et al., 2024) and use the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we follow the official implementation and use 50 inference steps.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been a foundation of computer vision. Classical methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) provide robust local descriptors for tasks ranging from object detection to image matching. Although these handcrafted features are effective at capturing local spatial patterns and geometric invariances, recent work in generative modeling has largely emphasized representations from modern self-supervised methods with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings support a complementary perspective: because the preservation of spatial structure is critical for generation quality, even classical spatial features can potentially improve diffusion training when they are properly aligned. This observation underscores the value of considering the full spectrum of spatial feature extractors, from traditional handcrafted descriptors to modern learned representations, provided that they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several roles: as discriminators that accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers for adversarial distillation in diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have improved convergence speed and enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation uses pretrained encoders to guide student networks toward higher-fidelity samples, producing clear perceptual-quality improvements. REPA (Yu et al., 2024), in particular, aligns diffusion features with external encoders and shows that representation alignment can improve both convergence and generation quality. Building on this line of work, we focus not merely on alignment itself, but on the preservation of spatial structure rather than on discriminative capability alone.

Denoising transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore

autoregressive formulations that combine denoising with sequential generation, enabling flexible trade-offs between quality and speed. Despite these architectural advances, training these models from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanisms behind this improvement remained unclear. Our analysis reveals that the key benefit comes from preserving spatial structure rather than semantic alignment, explaining why certain encoders provide stronger acceleration than others and guiding the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings complement this view by showing that when diffusion models are aligned with strong spatial representations from self-supervised encoders, both the generative and discriminative capabilities improve, suggesting that spatial structure preservation is a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work reveals a crucial insight: strong performance on discriminative benchmarks does not necessarily translate to better generation quality. Instead, we find that encoders preserving spatial structure, regardless of their classification accuracy—provide the most benefit for diffusion training. This suggests that the evaluation metrics for self-supervised encoders should be reconsidered when targeting generative applications, with spatial coherence being as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) to consider alternative formulations of the spatial-structure metrics discussed in §3. In addition, all figures in the paper are generated directly from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). We also use LLM assistance for searching and formulating relevant work in Appendix H, and we use Cursor in selected portions of the writing process.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) use linear probing to measure the representation gap, interpreting increases in validation accuracy as evidence for the effectiveness of REPA. We find that spatial-structure metrics (SSM) can likewise measure this representation gap and explain improvements across models. As shown in Figure 17, representation alignment helps close the gap between the SSM performance of a pretrained encoder such as DINOv2 and that of the diffusion features.

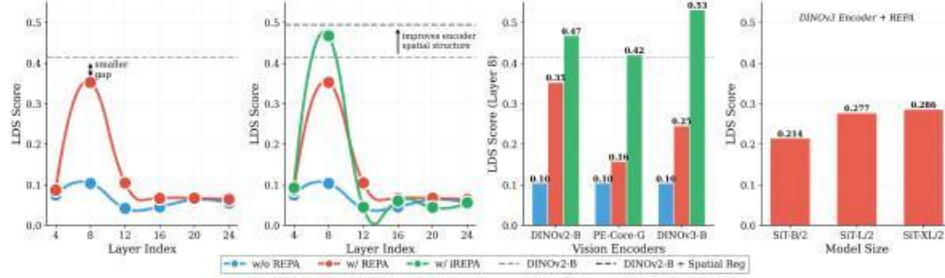


Figure 17: Representation alignment as bridging the spatial feature gap. a) Representation alignment can be interpreted as bridging the spatial-feature gap between diffusion features and vision-encoder patch features. b) iREPA (§4) accentuates the spatial features of the vision encoder, at the cost of some global information, thereby improving generation performance. c) iREPA consistently improves performance across vision encoders. d) Spatial-structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

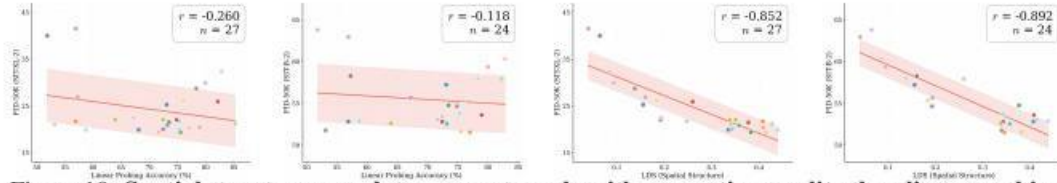


Figure 18: Spatial structure correlates more strongly with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear-probing accuracy versus FID for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$) exhibits weak correlation. **Right two:** Spatial structure (LDS) versus FID for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$) exhibits strong correlation. See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

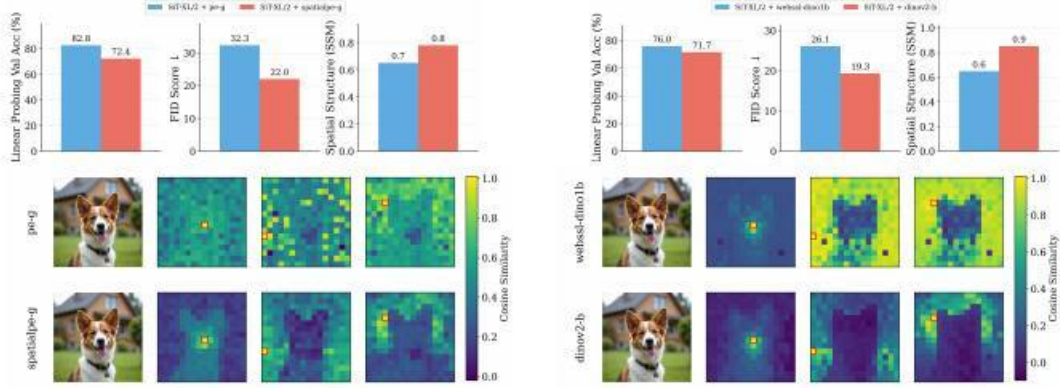


Figure 19: **Motivating anecdotes from recent SSL encoders: spatial structure matters.** **Top row:** Metric comparisons reveal an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g attains higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, accompanied by substantially weaker spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b likewise achieves higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, with weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine-similarity visualizations confirm that encoders yielding better generation (Spatial-PE-g, DINOv2-b) retain clear spatial coherence, whereas classification-oriented encoders (PE-g, WebSSL) lose spatial structure.

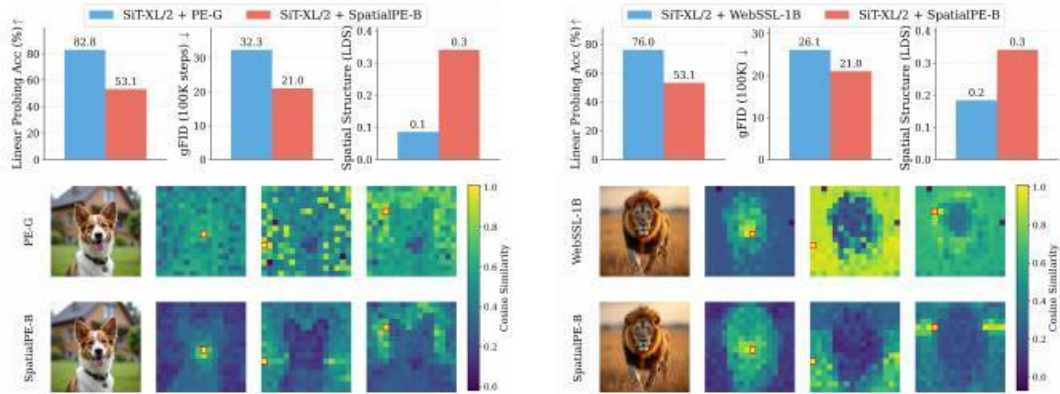


Figure 20: **Motivating examples: spatial structure matters.** **Top:** Metric comparisons reveal an inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite substantially higher validation accuracy (82.8% vs 53.1%), exhibits worse generation quality than PE_{Spatial}-B (Bolya et al., 2025). **Right:** Similarly, WebSSL-1B (Fan et al., 2025) has much stronger global performance (76.0% vs 53.1%) but worse generation. **Bottom:** Spatial structure provides a better predictor of generation quality than global performance. See §3 for the spatial-structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the effect of spatial normalization on vision-encoder representations. Vision encoders can contain substantial global components, or overlays, that limit contrast among spatial tokens. For example, tokens in one semantic region can become highly correlated with tokens in unrelated regions (e.g., background), thereby reducing the spatial distinctiveness of the features. Spatial normalization removes this global overlay and enhances contrast among spatial tokens, enabling the model to better preserve local spatial structure while reducing the dominance of global information that can interfere with generation quality.

The following figures illustrate this effect across several examples, showing how spatial normalization transforms feature representations so that spatial structure becomes more salient:

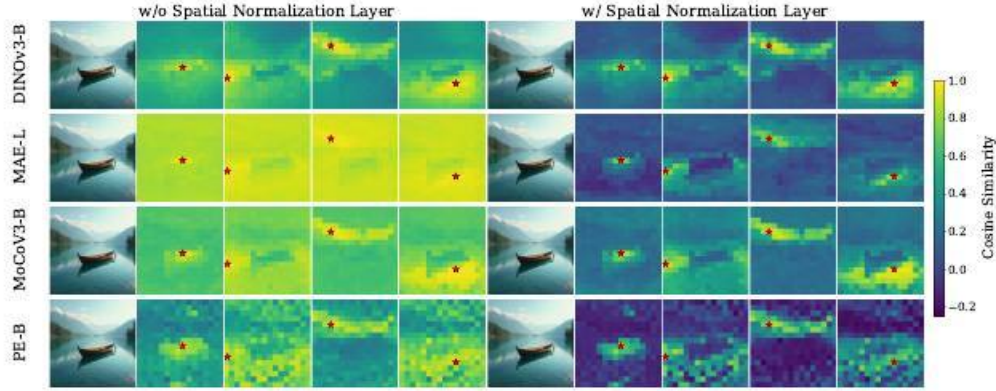


Figure 21: Visualizing the impact of spatial normalization (Example 1). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, producing more distinct semantic boundaries.

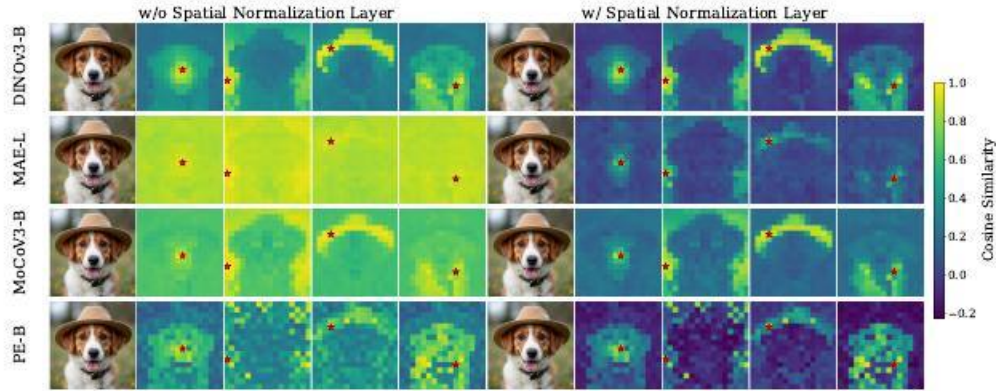


Figure 22: Visualizing the impact of spatial normalization (Example 3). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, producing more distinct semantic boundaries.

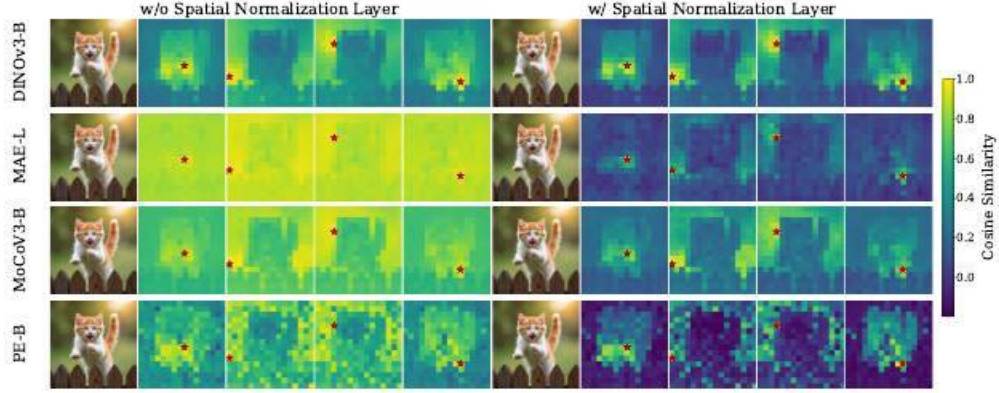


Figure 23: Visualizing the impact of spatial normalization (Example 2). Feature-similarity maps show how spatial normalization improves spatial contrast. The original features (left) exhibit a global overlay that reduces the distinction between foreground and background regions. After normalization (right), spatial tokens become more locally coherent, with clearer separation among semantic regions.

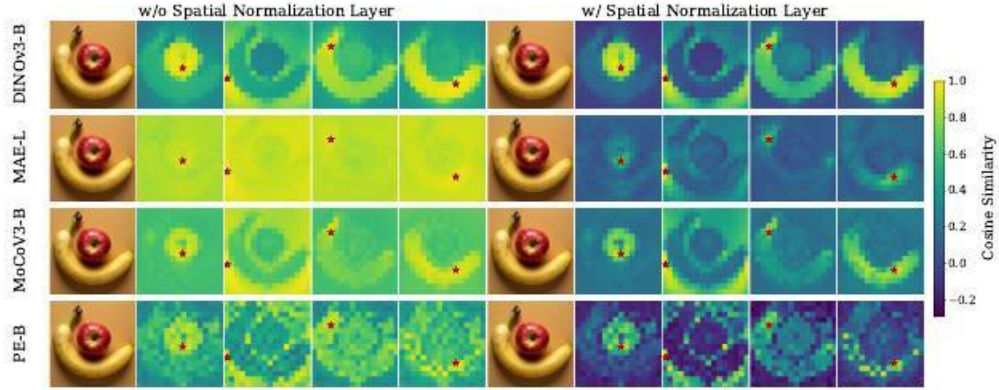


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps show token-similarity patterns before and after spatial normalization. Without normalization (left), global components induce high correlations across unrelated regions. With spatial normalization (right), local spatial structure is enhanced while global interference is reduced, producing more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct additional experiments on ImageNet-512 (Deng et al., 2009) to assess whether the proposed spatial improvements generalize to higher resolutions. We follow the REPA (Yu et al., 2024) setup and report results across several choices of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results use SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 reports REPA performance before and after applying the spatial improvements introduced by iREPA. We observe that these spatial improvements generalize to higher resolution and produce consistent gains across different pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

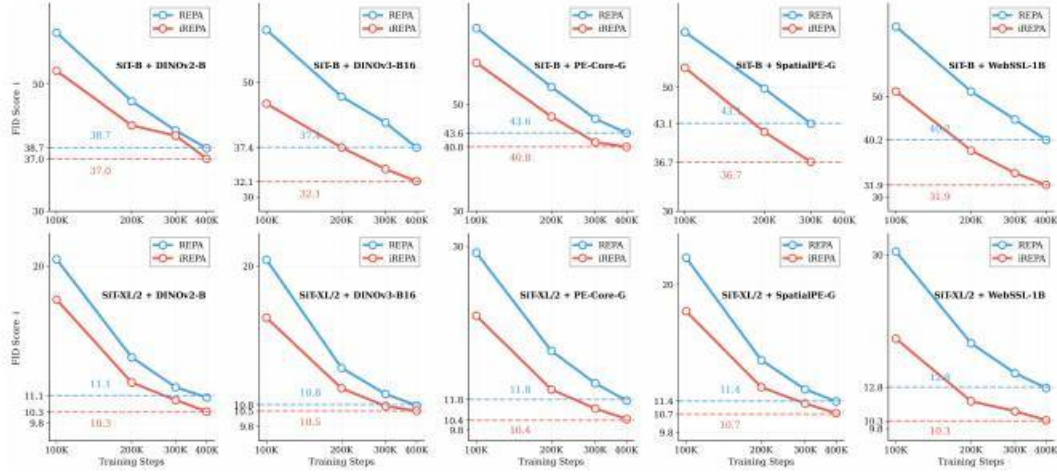


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed across resolutions, including both ImageNet 256 (Figure 7) and ImageNet 512 (shown here).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To further evaluate the generalizability of the proposed spatial improvements beyond ImageNet, we conduct extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we use MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with multiple pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 shows REPA results before and after applying the iREPA spatial improvements. The improvements generalize to multimodal T2I tasks and remain consistent across different pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

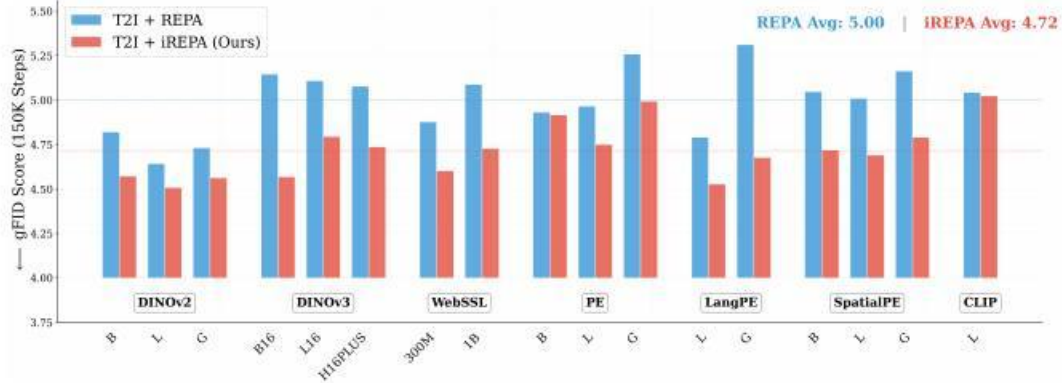


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both ImageNet (see §4) and multimodal T2I tasks (shown here). Consistent gains are also observed across different pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 presents the updated correlations across model sizes. Spatial structure continues to correlate much more strongly with generation performance than linear-probing accuracy. Interestingly, after outlier removal, linear probing exhibits a small positive correlation with gFID, meaning that generation becomes worse as linear-probing performance increases. This trend is consistent with the observations in Sec. 2, where target representations with higher global semantic performance often yield similar or worse generation performance under representation alignment (REPA).

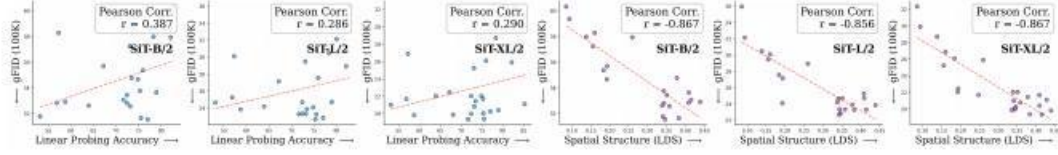


Figure 27: **Correlation analysis without outliers.** Across all model sizes (B, L, XL), spatial structure retains a much stronger correlation with generation performance (Pearson $|r| > 0.85$) than linear-probing accuracy (Pearson $|r| < 0.38$) after removing the outliers (MoCOv3-L and MAE-L). Interestingly, linear probing then shows a small positive correlation with gFID, indicating that generation worsens as linear-probing performance increases. This trend is consistent with Sec. 2, where target representations with higher global semantic performance often produce similar or worse generation performance under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after application of spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. Similar to quantitative improvements (§4), we also observe that spatial improvements (iREPA) also help improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024). Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We next verify the robustness of the key findings from both §3 and §4 with the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing. To study robustness of analysis from §3, we repeat the large-scale correlation analysis across different vision encoders with CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$) — demonstrating that key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD).

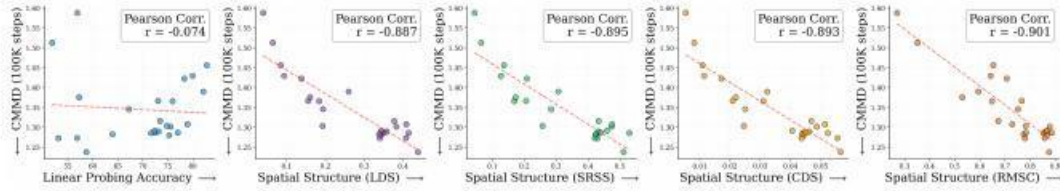


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 using CMMD (Jayasumana et al., 2024) instead of gFID. All results use SiT-B/2 (100K steps) and REPA. All spatial metrics correlate much more strongly with generation performance under CMMD (Pearson $|r| > 0.88$) than linear probing does (Pearson $|r| = 0.074$), demonstrating that the key empirical findings from §3 are robust to the choice of evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next assess whether the analysis in §4 generalizes when using CMMD (Jayasumana et al., 2024), in addition to traditional evaluation metrics (gFID, sFID, IS, etc.). Results are reported in Table 8. Across multiple vision encoders, accentuating the transfer of spatial features with iREPA improves convergence according to both CMMD and standard metrics (IS, FID, sFID, Prec., Recall). Consistent improvements are observed both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We report additional results using the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across various vision encoders, accentuating transfer of spatial features (iREPA) helps improve convergence speed with both CMMD and traditional evaluation metrics (IS, FID etc.). This demonstrates that key empirical findings from §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine the effect of spatial normalization (§4) on SAM2-S (46M) encoder features². As Table 9 shows, spatial normalization improves SAM2 performance, but the gains can be marginal. This behavior is expected because spatial normalization enhances spatial contrast by removing the global component, i.e., the mean of patch tokens. Since SAM2 already contains little to no global information (validation accuracy < 24%), spatial normalization produces only a modest increase in spatial contrast (see Figure 30).

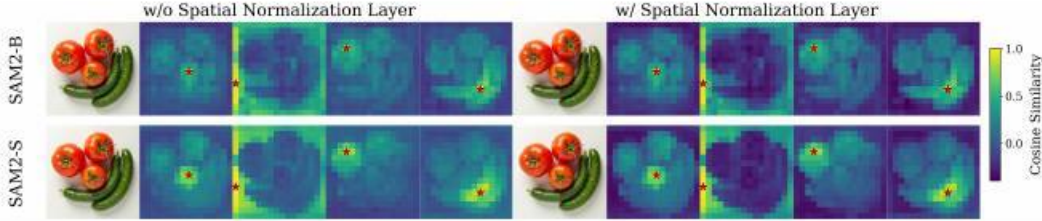


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial-normalization layer enhances spatial contrast, although the resulting improvements can be marginal. This is because spatial normalization (§4) enhances spatial contrast by removing the global component, i.e., the mean of patch tokens. Since SAM2 already contains little to no global information (validation accuracy < 24%), spatial normalization produces only a modest additional increase in spatial contrast.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results are reported with SiT-XL/2, without classifier-free guidance, using SAM2-S (46M) as the vision encoder and traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full fine-tuning³ rather than linear probing. Results are shown in Fig. 31. All results use SiT-XL/2 (100K steps) and REPA. To obtain full-finetuning validation accuracy, we fine-tune each vision encoder with learning rate $5e-5$, warmup ratio 0.1, and a total of 30 epochs. All spatial metrics correlate much more strongly with generation performance than full-finetuning accuracy. Interestingly, gFID shows a weak positive correlation with validation accuracy after full fine-tuning (Pearson $r = 0.317$), meaning that gFID increases, and generation performance worsens, as validation accuracy increases. This is consistent with the observations in §2, where target representations with higher global semantic performance often show similar or worse generation performance under representation alignment (REPA).

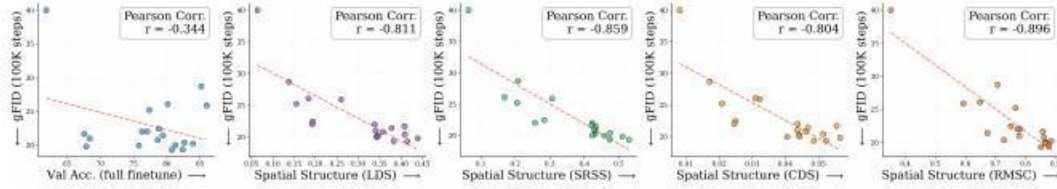


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 using validation accuracy after full fine-tuning instead of linear probing. All results use SiT-XL/2 (100K steps) and REPA. All spatial metrics correlate much more strongly with generation performance than full-finetuning accuracy. Interestingly, gFID shows a weak positive correlation with validation accuracy after full fine-tuning (Pearson $r = 0.317$), meaning that gFID increases, and generation performance worsens, as validation accuracy increases. This pattern is consistent with §2, where target representations with higher global semantic performance often yield similar or worse generation performance under representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Despite its simplicity, iREPA, when combined with JiT (Li & He, 2025a), outperforms recently proposed state-of-the-art pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results are reported after 200K training iterations with a batch size of 256 and are evaluated using 50-step Euler sampling without classifier-free guidance. DeCo results are taken directly from (Ma et al., 2025b).